INCH-POUND

MIL-I-85845A(AS)
14 November 1990
SUPERSEDING
MIL-I-85845(AS)
14 June 1988
MIL-I-22352/17(AS)
10 August 1984

MILITARY SPECIFICATION

INTERCOMMUNICATION SETS, AN/AIC-14A
MASTER INTERPHONE CONTROL PANEL C-2645D/AIC-14
GENERAL SPECIFICATION FOR

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance and acceptance requirements for C-2645D/AIC-14 Master Interphone Control Panel herein referred to as the C-2645D.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-636 Boxes, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5831

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SPECIFICATIONS (continued)

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-5087	Bonding, Electrical and Lighting Protection for Aerospace Systems
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-P-7788	Panels, Information Integrally Illuminated
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base NATO Code Number O-148
MIL-E-17555	Electronic and Electrical Equipment Accessories and Repair Parts; Packaging and Packing of
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-H-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft NATO Code Number H-537
MIL-I-85845/1	Intercommunication Sets, AN/AIC-14A Amplifier Module
MIL-I-85845/2	Intercommunication Sets, AN/AIC-14A VOX Module
MIL-I-85845/3	Intercommunication Sets, AN/AIC-14A Filter Module

STANDARDS

MILITARY

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference

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STANDARDS (continued)

MILITARY (continued)

MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Reliability Tests; Exponential Distribution
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-882	System Safety Program for Systems and Associated Subsystems and Equipment, Requirements for
MIL-STD-965	Parts Control Program
MIL-STD-45662	Calibration Systems Requirements

2.1.2 Other Government documents, drawings and publications. The following documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DOCUMENTS

NAVAL AIR SYSTEMS COMMAND

AD-1115	Electromagnetic Compatibility Design Guide for Avionics and Related Ground Support Equipment

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

DL 1794AS7000	Panel, Interphone Control, Master, C-2645D/AIC-14

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building No. 4, Section D, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for the associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Item description. The C-2645D shall be capable of providing the following:

a. Amplification of incoming interphone and receiver audio.

b. Selection of the mode of operation of the associated microphone.

c. Amplification of microphone audio for outgoing interphone signals.

d. Two identical audio amplifiers each capable of operation as an interphone amplifier or isolation amplifier.

e. Capability to operate with degraded equipment in failure modes involving single amplifier operation or alternatively, by-passing the amplifiers completely.

f. Current supply to power a microphone preamplifier or carbon microphone.

g. Voice actuated keyswitch and connection of microphone generated interphone signals to the interphone output.

An outline drawing of the C-2645D is provided on Figure 1.

3.3 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2 and 6.3).

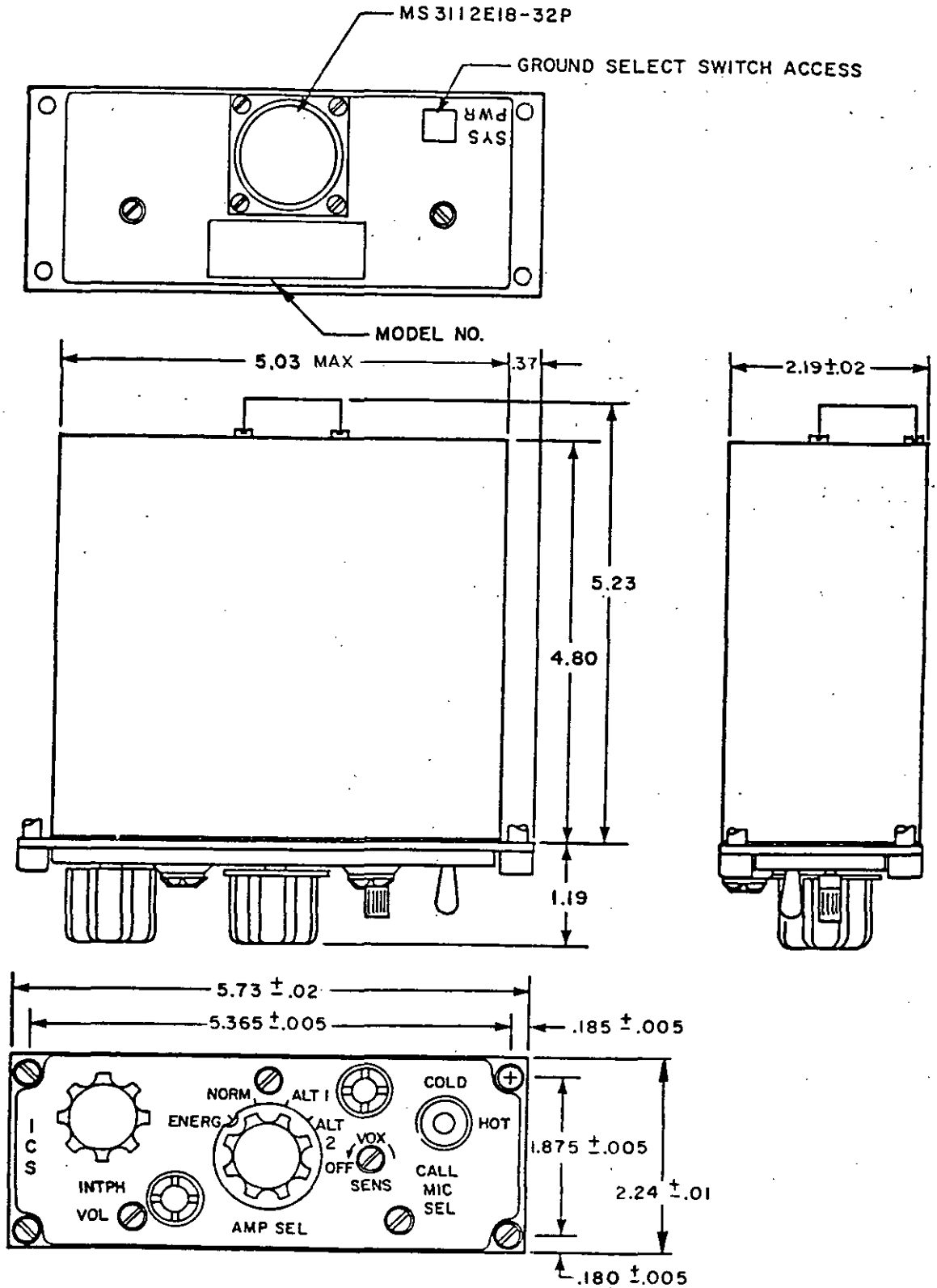
3.4 Parts and materials.

3.4.1 Nonstandard parts and material approval. Approval for the use of nonstandard parts and materials shall be obtained by the contractor as specified in MIL-STD-965, Procedure I.

3.4.2 Parts rescreening. The contractor shall provide 100 percent rescreening of active components such as but not limited to ICs, hybrids, semiconductors, and other components, as appropriate to provide a quality product. Active component parts rescreening at the electronic equipment manufacturer's plant shall consist of functional check of critical parameters at +125°C, ambient, and -54°C to the part specification.

3.4.3 Supplemental requirements. In the event that the contractor proposes to alter the materials or piece parts of the C-2645D specified herein, the supplemental requirements of 3.4.2.1 through 3.4.2.3 shall be applicable for the selection of parts and materials.

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

FIGURE 1. C-2645D outline.

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3.4.3 Supplemental requirements. In the event that the contractor proposes to alter the materials or piece parts of the C-2645D specified herein, the supplemental requirements of 3.4.2.1 through 3.4.2.3 shall be applicable for the selection of parts and materials.

3.4.3.1 Basic conformance. Parts and materials selected for use in the C-2645D shall be in accordance with MIL-E-5400, unless the parts or materials listed therein will not perform adequately in the intended applications and environments specified herein. When any parts and materials not in accordance with MIL-E-5400 must be used, refer to 3.4.1.

3.4.3.2 Toxic and corrosive fumes. The materials, as used in the manufacture of the C-2645D and under the conditions therein, shall not liberate deleterious or corrosive fumes. This shall include fungicidal agents that are used.

3.4.3.3 Recycled, virgin and reclaimed materials. Except when intended use of the C-2645D will be jeopardized by the use of reclaimed or recycled materials, these materials shall be used to the maximum extent possible.

3.5 Design and construction. Design and construction of the C-2645D shall conform with MIL-E-5400, DL 1794AS7000 and as specified herein. It is a condition of final acceptance that the C-2645D manufactured shall meet all the design, construction and workmanship requirements of this specification and applicable acceptance test documents.

3.5.1 Reliability.

3.5.1.1 Operational stability. The C-2645D shall operate with required performance, continuously or intermittently without the necessity of readjustment of controls which are inaccessible during flight.

3.5.1.2 Reliability in mean-time-between-failure (MTBF). The lower test MTBF θ_1 for the C-2645D shall be no less than 1000 hours.

3.5.1.3 Operating life. The C-2645D shall have a minimum total operating life of not less than 10,000 hours when the maintenance delineated in the maintenance plan is performed in accordance with approved procedures.

3.5.2 Interchangeability. Interchangeability shall exist between all C-2645Ds and all replaceable assemblies, subassemblies and parts of the C-2645Ds in accordance with MIL-E-5400. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance without need for mechanical or electrical matching, alignment or adjustment of parts or circuitry.

3.5.3 Transportability. The C-2645D shall be transportable by air, rail, truck or ship when packaged as specified in Section 5.

3.5.4 Dimensions. The dimensions of the C-2645D shall be as listed below with a tolerance of ±0.03 inches (see Figure 1):

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a. Height 2.24 ± .01 inches (at front plate).

b. Width 5.73 ± .02 inches (at front plate).

c. Depth 4.80 inches (beyond front plate, not including connector and control protrusions).

3.5.5 Weight. The weight of the C-2645D shall be not greater than 3.0 pounds.

3.5.6 Color. The color of the C-2645D shall be as identified in DL 1794AS7000.

3.5.7 Connectors. The input and output connectors shall be as identified in DL 1794AS7000.

3.5.8 Nameplates and identification marking. Serial number assignment and nameplate approval shall be in accordance with MIL-N-18307. Identification marking shall be in accordance with MIL-STD-130. The following information shall be placed on the nameplate:

a. Unit nomenclature (Panel, Interphone Control, Master).

b. Unit designator (C-2645D/AIC-14A).

c. Contract number.

d. Manufacturer's code number.

e. Equipment drawing number (30003 - 1794AS7000).

f. Technical manual identification.

g. Procuring activity (Department of the Navy - Naval Air Systems Command).

h. Serial number specified in the contract shall be stamped on the C-2645D.

3.5.8.1 Serial numbers. Serial numbers shall be assigned in accordance with MIL-N-18307 for the C-2645Ds under contract or purchase order. The contractor shall serialize sequentially all C-2645Ds which are delivered under the contract.

3.5.9 Safety. The contractor shall ensure that safety requirements are in accordance with MIL-STD-882.

3.5.10 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

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a.	Temperature	Room ambient (25 ± 10°C).
b.	Altitude	Normal ground (0 - 5000 feet).
c.	Vibration	None.
d.	Humidity	Normal ambient up to 90 percent relative humidity.
e.	Input power	28 Vdc, 20 amperes.

3.5.11 Service conditions. The C-2645D shall operate as required under any of the environmental service conditions, or any probable combination thereof, as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

3.5.11.1 Altitude. The C-2645D shall operate as required at any pressure up to 50,000 feet.

3.5.11.2 Fungus. The C-2645D shall withstand exposure to fungal growth. Overspraying of the C-2645D shall be prohibited.

3.5.11.3 Fluids. The C-2645D shall be unaffected by contact with the following fluids:

a. JP-4 and JP-5 fuels (see MIL-T-5624).

b. Hydraulic fluids (see MIL-H-83282).

c. Lubricating oil (see MIL-L-7808).

d. Coolants of the fluorocarbon, silicone, silicate ester and glycol families.

3.5.12 Warm-up time. The time required for the C-2645D to warm up prior to providing voice communications shall be not greater than 10 seconds under standard conditions and 30 seconds under service conditions. The time required for the C-2645D to warm up in order to meet the performance parameters specified herein shall be not greater than 2 minutes under standard conditions and 4 minutes under service conditions.

3.5.13 Input power. The C-2645D shall meet all applicable requirements of MIL-STD-704 and shall meet specified performance requirements when energized from a power source having characteristics and limits specified in MIL-STD-704 for 28 volts dc.

3.5.13.1 Steady state power.

3.5.13.1.1 Normal steady state. The C-2645D shall meet the requirements of MIL-STD-704 for normal steady state dc input power.

3.5.13.1.2 Emergency steady state. The C-2645D shall meet the requirements of MIL-STD-704 for emergency steady state dc input power.

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3.5.13.2 Transient.

3.5.13.2.1 Voltage surge. The C-2645D shall meet the requirements of MIL-STD-704 for voltage surges. The C-2645D shall not be failed for degraded performance during the voltage surges if the C-2645D returns to normal operation without damage after voltage surges pass.

3.5.13.3 Overvoltage, undervoltage and power interruption.

3.5.13.3.1 Overvoltage. The C-2645D shall meet the requirements of MIL-STD-704 for overvoltage. The C-2645D shall not be failed for degraded performance during overvoltage exposure if the C-2645D returns to normal operation without damage after the exposure.

3.5.13.3.2 Undervoltage. The C-2645D shall meet the requirements of MIL-STD-704 for undervoltage. The C-2645D shall not be failed for degraded performance during undervoltage exposure if the C-2645D returns to normal operation without damage after the exposure.

3.5.13.3.3 Power interruption. The C-2645D shall meet the requirements of MIL-STD-704 for power interruption. The C-2645D shall not be failed for degraded performance during power interruption if the C-2645D returns to normal operation without damage after the exposure.

3.5.14 Electromagnetic interference (EMI). The C-2645D shall comply with the requirements of MIL-STD-461, Class A1b equipment, as modified herein. These requirements shall be met in all operating modes and at all operating frequencies of the C-2645D at standard conditions specified in 3.5.10. EMI testing shall be accomplished in accordance with the procedures of MIL-STD-462 to demonstrate compliance with requirements of MIL-STD-461. EMI design guidelines shall be in accordance with AD-1115.

3.5.14.1 Conducted and radiated emissions.

3.5.14.1.1 Conducted emission, power and interconnecting leads. Conducted emissions from the C-2645D power and interconnecting leads shall be within the limits of MIL-STD-461, Requirement CE03.

3.5.14.1.2 Conducted emissions, power leads, spikes, time domain. Conducted emissions from the C-2645D power leads shall be within the limits of MIL-STD-461, Requirement CE07.

3.5.14.1.3 Radiated emissions electric field 14 kHz to 10 GHz. Radiated emissions from the C-2645D, electric field, between 14 kHz and 10 GHz, shall be within the limits of MIL-STD-461, Requirement RE02. Above 30 MHz, the limits shall be met for both horizontally and vertically polarized waves.

3.5.14.2 Conducted and radiated susceptibility.

3.5.14.2.1 Conducted susceptibility, power leads, 30 Hz to 50 kHz. The C-2645D shall not exhibit any of the indications of susceptibility specified in 3.5.14.3 when exposed to electromagnetic energy injected onto its power leads less than or equal to the values of MIL-STD-461, Requirement CS01.

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3.5.14.2.2 Conducted susceptibility, power leads, 50 kHz to 400 MHz. The C-2645D shall not exhibit any of the indications of susceptibility specified in 3.5.14.3 when exposed to 1-volt from a 50-ohm source across a calibrating resistor as specified in MIL-STD-461, Requirement CS02. This requirement shall apply to all neutrals and grounds whether or not they are internally returned to chassis ground.

3.5.14.2.3 Conducted susceptibility, spikes, power leads. The C-2645D shall not exhibit any of the indications of susceptibility specified in 3.5.14.3 when spikes each having the waveform specified in MIL-STD-461, Requirement CS06, are applied to the ac and dc power input leads.

3.5.14.2.4 Radiated susceptibility, magnetic induction field, spikes and power frequencies. The C-2645D shall not exhibit any of the indications of susceptibility specified in 3.5.14.3 when exposed to radiated signals specified in MIL-STD-461, Requirement RS02.

3.5.14.2.5 Radiated susceptibility, electric field, 14 kHz to 40 GHz. The C-2645D shall not exhibit any of the indications of susceptibility specified in 3.5.14.3 when exposed to the radiated electric fields specified in MIL-STD-461, Requirement RS03, except field levels and modulation shall be as specified in Table I.

3.5.14.3 Susceptibility interpretation. The criteria of 3.5.14.3.1 and 3.5.14.3.2 shall be applied to determine whether the unit is susceptible.

3.5.14.3.1 Susceptibility to stray magnetic fields. The C-2645D shall be shielded or so designed that when oriented in any position within a magnetic field of 2 Gauss in the frequency range of 400 Hz to 2800 Hz. The properly terminated output from each amplifier circuit shall not be greater than 10.0 millivolts at 400 Hz and shall be not greater than a linear increase of 16 uV/Hz up to 2800 Hz.

3.5.14.3.2 Susceptibility. The C-2645D shall be designed to resist audio interference including power supply pickup. The C-2645D shall exhibit audio outputs not greater than -70 dB within the audio pass band of 100 Hz to 10 kHz when subjected to the radiated and conducted signal exposures required herein.

3.5.14.4 Ground plane interference. The C-2645D shall be capable of specified performance when subjected to aircraft chassis noise between the interfacing WRAs.

3.5.14.5 Grounding requirements. The grounding system between WRAs of the equipment shall be compatible and consistent with the following aircraft grounding scheme. All WRAs shall be grounded in such a manner as to prevent ground loops and common ground returns for signal and power circuits, minimize EMI, and to protect personnel from electrical hazards.

3.5.14.6 Bonding. A means of electrically bonding the equipment to the aircraft chassis shall be provided and shall comply with the requirements of MIL-B-5087. Bonding resistance between the bonding surface and all parts of the equipment case shall be not greater than 2.5 milliohms.

3.5.15 Interface.

3.5.15.1 Signal interface. The C-2645D input and output signals shall have the characteristics specified in Tables II and III.

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TABLE I. RS03 signal (continuous application test levels).

Frequency (MHz)	Modulation				Power Density (mW/cm^2)		v/M(Ref) 6/
					Peak 5/	Ave	
0.014-850	AM 90% 1 kHz Sine Wave				--	0.66	50
215-225	PAM 1/	20 usec 2/	PW 3/	200 pps 4/	0.126	--	22
215-225	PAM	200 usec	PW	200 pps	0.126	--	22
400-450	PAM	4 usec	PW	300 pps	0.200	--	28
850-940	PAM	10 usec	PW	1000 pps	0.316	--	35
850-940	PAM	125 usec	PW	300 pps	0.316	--	35
940-1215	PAM	10 usec	PW	1000 pps	0.106	--	20
1215-1365	PAM	10 usec	PW	3000 pps	0.106	--	20
1365-2900	PAM	1 usec	PW	1000 pps	0.160	--	25
2900-3100	PAM	3 usec	PW	1000 pps	1.60	--	78
2900-3100	PAM	70 usec	PW	200 pps	1.60	--	78
3100-3500	PAM	10 usec	PW	3000 pps	3.18	--	110
3100-3500	PAM	50 usec	PW	3000 pps	3.18	--	110
3500-9000	PAM	1 usec	PW	1000 pps	0.636	--	50
9000-14000	PAM	0.2 usec	PW	1000 pps	0.106	--	20
14000-18000	PAM	0.2 usec	PW	50000 pps	0.106	--	20
33200	PAM	0.2 usec	PW	2000 pps	0.106	--	20

NOTES:
1/ Pulse amplitude modulated.
2/ Microseconds.
3/ Pulse width.
4/ Pulses per second.
5/ Peak power density for PAM signals is specified as the power density measured by an average power measuring device if the generator is run CW; for example, average power density within the pulse envelope.
6/ Volts per meter equivalent is shown for reference only.
7/ Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations and field strength shown.

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3.5.15.1.1 Grounds.

3.5.15.1.1.1 Power ground (PWR GND). Primary power shall be specified as power delivered to the C-2645D by the aircraft generating and distribution system. Primary power returns shall be connected to the chassis of the C-2645D.

3.5.15.1.1.2 System ground (SYS GND). The system ground network shall be established to provide isolation from interference injected onto the power returns. The interconnection of signal returns form a system ground network. In the C-2645D, system ground shall be isolated from power ground except as modified in 3.5.15.2.1.5.

3.5.15.1.1.3 Microphone return (MIC RTN). The signal return for the microphone shall be system ground.

3.5.15.1.1.4 Headset return (HDST RTN). The signal return for the headset shall be system ground.

3.5.15.2 Operator interface.

3.5.15.2.1 Controls. The C-2645D shall be provided with operation controls as specified in 3.5.15.2.1.1 through 3.5.15.2.1.5.

3.5.15.2.1.1 Amplifier selector (AMP SEL). The AMPL SEL switch shall be a rotary four-position switch which shall determine the modes of operation for the interphone and isolation amplifiers as specified in 3.5.15.2.1.1.1 through 3.5.15.2.1.1.4.

3.5.15.2.1.1.1 Emergency position (EMERG). The emergency mode of operation is available to allow operation of radio receivers and transmitters in the event of a loss of primary power or total failure of some critical ICS assembly. When the emergency mode is selected, direct communication with other ICS stations is not available. The sidetone audio from the transmitter may be monitored by other ICS stations, permitting indirect station to station communications. In the EMERG position, the interphone amplifier shall be electrically disconnected from the system. Audio from the associated microphone, or microphone circuit, shall be routed to the input of a selected transmitter, and the emergency radio receiver outputs shall be connected directly to the associated headset.

3.5.15.2.1.1.2 Normal position (NORM). In the NORM position, the interphone and isolation amplifiers shall function as follows:

a. The interphone amplifier shall amplify the audio signal derived from the associated microphone, or microphone circuit, to a level sufficient both for use on the outgoing interphone line and for the use as sidetone signal input to the isolation amplifier.

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TABLE II. Input signals.

Signal Name	Characteristics
Interphone (INTPH)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0.01 to 1.0 Vac.
Microphone Audio (MIC AUDIO)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0.1 to 1.0 Vac.
Split Interphone (SPLIT)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0.01 to 1.0 Vac. No dc component normally present.
Auxiliary Interphone (AUX INTPH)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0.01 to 1.0 Vac.
Select Receiver (SEL RCVR)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from .01 to 1.0 Vac.
Auxiliary CALL (AUX CALL)	Relay logic signal: (1) Open; (2) Short to ground.
Emergency Receiver No. 1 (EMER RCVR NO. 1)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 15 Vac.
Emergency Receiver No. 2 (EMER RCVR NO. 2)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 15 Vac.
COLD Interphone Key (COLD INTPH KEY)	Relay logic signal: (1) Open; (2) Short to ground.
External Transmit Key (EXT XMIT KEY)	Relay logic signal: (1) Open; (2) Short to ground.
Interphone Control (INTPH CONTROL)	Relay logic signal: (1) Open; (2) Short to ground.
Power (PWR)	+27.5 $\pm$ 0.5 Vdc.
Panel Lights (LIGHTS)	0 to +28 Vdc or 0 to 6 Vac, 400 Hz.

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TABLE III. Output signals.

Signal Name	Characteristics
CALL, Emergency Override (CALL)	Relay logic signal: (1) Open; (2) Short to ground.
Transmitter Key (XMIT KEY)	Relay logic signal: (1) Open; (2) Short to ground.
Emergency Receiver No. 3 (EMER RCVR NO. 3)	Relay logic signal: (1) Open; (2) Short to ground.
Split Interphone (SPLIT)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from .01 to 27 Vac.
Transmit, Audio (XMIT AUDIO)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0.01 to 1.0 Vac.
Interphone (INTPH)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0.01 to 27 Vac.
Low Z Headset (HDST LO Z)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 3.0 Vac.
Audio (AUDIO)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 27 Vac.
Auxiliary Mic Current (AUX MIC CUR)	60 $\pm$ 10 milliamperes.
Filtered Power (FILT PWR)	+25.5 to +28 Vdc.

b. The isolation amplifier shall amplify signals derived from radio receivers, incoming interphone lines, and sidetone output of the interphone amplifier, delivering the audio output to the associated headset or headset circuit.

Provisions for split headset operation shall also be incorporated in the interphone control unit. During split headset operation with AMP SEL switch in the NORM position, and the microphone select switch in the cold position, the interphone amplifier shall act as a second isolation amplifier with the same capabilities as the normal isolation amplifier except that the input signals and the output load shall be separate and independent. For a discussion of the interphone relay function including split headset operation (see 3.5.15.2.3.3).

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3.5.15.2.1.1.3 Alternate one position (ALT 1). In the ALT 1 position, the interphone amplifier output shall be electrically disconnected from the system and the isolation amplifier shall accomplish both amplifier functions on a time-share basis (that is act as an interphone amplifier for ICS communications, and as an isolation amplifier at all other times). The overall change in gain in either the interphone or isolation amplifier modes resulting from this type of operation shall be not greater than ±4 dB from gain characteristics during normal operation.

3.5.15.2.1.1.4 Alternate two position (ALT 2). In the ALT 2 position, the isolation amplifier output shall be electrically disconnected from the system and the interphone amplifier shall accomplish both amplifier functions on a time-share basis (that is act as an interphone amplifier for ICS communications, and as an isolation amplifier at all other times). The overall changes in gain in either the interphone or isolation amplifier modes resulting from this type of operation shall be not greater than ±4 dB from gain characteristics during normal operation.

3.5.15.2.1.2 Microphone selector switch (MIC SEL). The MIC SEL switch shall be a three-position toggle switch. Position of this switch shall determine the mode of interphone transmission initiation. The three initiation modes are as specified in 3.5.15.2.1.2.1 through 3.5.15.2.1.2.3.

3.5.15.2.1.2.1 COLD. The COLD position shall be a fixed switch position. It shall provide for interphone transmission initiation predicated upon the state of an external interphone push-to-talk switch.

3.5.15.2.1.2.2 HOT. The HOT position shall be a fixed switch position. It shall provide for continuous transmission of interphone audio. This position shall be used in conjunction with the VOX circuitry (see MIL-I-85845/2) to provide interphone transmissions initiated by the operator's voice.

3.5.15.2.1.2.3 CALL. The MIC SEL switch shall have a spring loaded CALL position. Upon release, the MIC SEL switch shall return to the HOT position. In the CALL position, the operation of the C-2645D shall be the same as specified above for the HOT position except that all interconnected ICS stations shall be activated to receive the outgoing interphone signal generated by this unit.

3.5.15.2.1.3 Interphone volume control (INTPH VOL). The interphone volume control shall be a rotary type control, providing an adjustment range of 35 ± 3 dB. The amplitude of incoming radio receiver signals shall not be affected by more than ±1 dB when the INTPH VOL control is adjusted throughout its range.

3.5.15.2.1.4 VOX sensitivity control (VOX SENS). VOX SENS controls selection of and input sensitivity for voice operated transmission. With the MIC SEL switch in the HOT position and VOX SENS rotated fully counterclockwise into the detent position, the VOX circuit is bypassed and the unit is in a continuous "Key ON" condition. When VOX SENS is not in its detent position, interphone transmission is triggered by operator voice level at the microphone (see MIL-I-85845/2). The VOX SENS control shall adjust the MIC AUDIO signal

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amplitude to the VOX module. Trigger sensitivity to the MIC AUDIO signal shall be variable from 20 mVAC maximum in the full clockwise position through 4.0 VAC minimum in the counterclockwise position. In the midrange position, trigger sensitivity shall be 180 + 50 mVAC. Conditions which shall disable the VOX function include selection of COLD position of the MIC SEL switch, selection of ALT 1 or ALT 2 on the AMP SEL switch, or initiation of a radio transmission.

3.5.15.2.1.5 Ground select (GND SEL). The C-2645D shall have an internally mounted double pole two-position toggle switch having access for selection through a cutout in the rear of the unit. In the power position, the system ground shall be electrically connected to the power ground. In the system position, the two grounds shall be electrically isolated. Also the EMER RCVR NO. 3 signal shall be routed through the switch (see 3.6.1.9).

3.5.15.2.2 Panel lighting. The C-2645D shall be provided with two edge lighting panel lamps to enhance night vision. The panel lamps shall be mounted in accordance with DL 1794AS7000. Input power may range from 0 to +28 Vdc or 6 Vac at 400 Hz. Accommodation for this power variation shall be made by user selection of lamps from Table IV. The C-2645D shall be capable of operating with either ac or dc lighting power source as permitted in MIL-C-7681. When ac power is used for panel lighting, self susceptibility to 400 Hz power source shall induce not more than 5 millivolts across any properly terminated output with the input circuits grounded.

TABLE IV. Front panel bulbs.

Nomenclature	Power
MS25237A-327	+28 Vdc
MS25237A-328	6 Vac

3.5.15.2.3 Relay functions. The C-2645D shall provide the relay functions specified in 3.5.15.2.3.1 through 3.5.15.2.3.3.

3.5.15.2.3.1 Emergency relay. When EMERG is selected on the AMPL SEL switch, there are no provisions for direct station to station ICS communications. The emergency relay shall provide switching of the MIC AUDIO and EXT XMIT KEY inputs so that they are connected to the XMIT AUDIO and XMIT KEY outputs, respectively. These outputs, after passing through other station WRAs, drive transmitters whose sidetone audio may be monitored by other ICS stations, permitting indirect station to station communications.

3.5.15.2.3.2 Key relay. The key relay is used to connect the MIC AUDIO and EXT XMIT KEY inputs to the XMIT AUDIO and XMIT KEY outputs when the microphone key switch is closed and the C-2645D is operating in NORM, ALT 1 or ALT 2.

3.5.15.2.3.3 Interphone relay. The interphone relay shall select between the interphone operation and split headset operation. In interphone operation, the C-2645D shall provide output INTPH and AUDIO signals derived from MIC AUDIO, INTPH, AUX INTPH and SEL RCVR inputs. In split headset

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operation, the C-2645D shall provide the SPLIT audio output signal derived from the SPLIT input.

3.6 Performance. The performance of the C-2645D shall be as specified in 3.6.1 through 3.6.2.7.8 and Drawing List 1794AS7000. Unless otherwise specified herein, these requirements shall apply under both standard and service conditions (see 3.5.10 and 3.5.11).

3.6.1 Input characteristics. The input signals listed in Table II shall have the characteristics as specified in 3.6.1.1 through 3.6.1.14.

3.6.1.1 Interphone (INTPH).

3.6.1.1.1 Signal frequency (INTPH). The interphone signal input shall accept input drive signals which range from 300 to 7000 Hz.

3.6.1.1.2 Signal amplitude (INTPH). The INTPH signal input shall accept input drive signals which range in amplitude from 0.01 to 1.0 Vac.

3.6.1.1.3 Input impedance (INTPH). The impedance of the INTPH input shall be 660 ohms ± 10 percent with the AMP SELECT switch in the NORM position and INTPH VOL fully clockwise.

3.6.1.2 Mic audio (MIC AUDIO).

3.6.1.2.1 Signal frequency (MIC AUDIO). The MIC AUDIO signal input shall accept input drive signals which range from 300 to 7000 Hz.

3.6.1.2.2 Signal amplitude (MIC AUDIO). The MIC AUDIO signal input shall accept input drive signals which range in amplitude from 0.1 and 1.0 Vac. The circuitry shall be capable of withstanding overvoltage inputs of up to 2.0 Vac without sustaining damage.

3.6.1.2.3 Input impedance (MIC AUDIO). The impedance of the MIC AUDIO signal input shall be 100 ohms ± 10 percent.

3.6.1.2.4 Microphone current. The MIC AUDIO input line shall be capable of supplying 50 ± 10 milliamperes to the microphone circuit for carbon microphones having an impedance of 100 ohms or as a source of power to equivalent dynamic microphone and associated preamplifier combinations.

3.6.1.3 Split interphone (SPLIT).

3.6.1.3.1 Signal frequency (SPLIT). The SPLIT signal input shall accept input drive signals which range from 300 to 7000 Hz.

3.6.1.3.2 Signal amplitude (SPLIT). The SPLIT signal input shall accept input drive signals which range in amplitude from 0.1 to 1.0 Vac.

3.6.1.3.3 Input impedance (SPLIT). The impedance of the SPLIT signal input shall be 1200 ohms ± 10 percent.

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3.6.1.4 Auxiliary interphone (AUX INTPH). The input characteristics of the AUX INTPH signal input shall be the same as specified for the INTPH signal input of 3.6.1.1.

3.6.1.5 Select receiver (SEL RCVR). The SEL RCVR input shall be applied directly to the amplifier (see MIL-I-85845/1) which acts as an isolation amplifier. Also, the signal shall be routed to the interphone amplifier acting as the isolation amplifier when ALT 2 has been selected on the AMPL SEL control.

3.6.1.6 Auxiliary CALL (AUX CALL). The AUX CALL input shall be a relay logic signal having the following two states:

a. Open.

b. Short to ground.

The AUX CALL signal shall be routed to the MIC SEL switch.

3.6.1.7 Emergency receiver No. 1 (EMER RCVR NO. 1). The EMER RCVR NO. 1 signal shall be routed to the AUDIO output via the AMPL SEL switch when the EMERG position is selected. The EMER RCVR NO. 1 signal shall be supplied by the R/T Interphone Control panel and will be the sum of three radio receive channel outputs. The signal shall have the characteristics specified in 3.6.1.7.1 and 3.6.1.7.2.

3.6.1.7.1 Signal frequency (EMER RCVR NO. 1). The EMER RCVR NO. 1 signal input shall accept input drive signals which range from 300 to 7000 Hz.

3.6.1.7.2 Signal amplitude (EMER RCVR NO. 1). The EMER RCVR NO. 1 signal input shall accept input drive signals which range in amplitude from 0 to 15 Vac.

3.6.1.8 Emergency receiver No. 2 (EMER RCVR NO. 2). The EMER RCVR NO. 2 signal shall be routed to the AUDIO output via the AMP SEL switch when the EMERG position is selected. EMER RCVR NO. 2 signal characteristics are the same as those of EMER RCVR NO. 1.

3.6.1.9 Emergency receiver No. 3 (EMER RCVR NO. 3). The EMER RCVR NO. 3 signal shall be routed as follows:

a. When the EMERG position is selected on the AMP SEL switch and the rear mounted GND SEL switch of 3.5.15.2.1.5 is in the PWR GND position, the EMER RCVR No. 3 signal shall be an open condition. The EMER RCVR NO. 3 signal is routed to the AUDIO output via the AMP SEL switch.

b. When the EMERG position is selected on the AMP SEL switch and the rear mounted GND SEL switch of 3.5.15.2.1.5 is in the SYS GND position, the EMER RCVR NO. 3 signal shall be an open condition.

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3.6.1.10 COLD interphone key (COLD INTPH KEY). The COLD INTPH KEY input shall be a relay logic signal having the following two states:

a. Open.

b. Short to ground.

The COLD INTPH KEY Input shall be routed to the MIC SEL switch.

3.6.1.11 External transmit key (EXT XMIT KEY). The EXT XMIT KEY signal is a relay logic drive signal having the following two states based on the position of the operator's keyswitch:

a. Open.

b. Short to ground.

The signal shall be applied to the input circuitry to activate the keyswitch relay function. The signal shall also disable the VOX function with a short to ground.

3.6.1.12 Interphone control (INTPH CONTROL). The INTPH CONTROL signal is a relay logic drive signal having the following two states:

a. Open.

b. Short to ground.

The signal shall be applied to actuate the interphone relay circuitry as necessary to comply with the performance specified herein.

3.6.1.13 Power (PWR). The C-2645D shall accept 27.5 + 0.5 Vdc power. In NORMAL operation, no load on the +27.5 Vdc filtered power output (see 3.6.2.7.8) and the outputs driven to rated power, the input current shall be not greater than 600 ma.

3.6.1.14 Panel lights (LIGHTS). The C-2645D shall be capable of accepting either 0 to +28 Vdc or 0 to 6 Vac 400 Hz power to illuminate the two front panel lamps.

3.6.2 Output characteristics. The output signals listed in Table II shall have characteristics specified in 3.6.2.1 through 3.6.2.7.8.

3.6.2.1 CALL, emergency override (CALL). The CALL output shall have the following two states:

a. Open.

b. Short to ground.

The CALL output shall be routed to the MIC SEL switch.

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3.6.2.2 Key transmitter (XMIT KEY). The XMIT KEY output shall be a logic signal having the following two states:

a. Open.

b. Short to ground.

The XMIT KEY signal shall be a short to ground when transmitter keying is desired. Transmitter keying is commanded by actuation of the microphone key-switch resulting in a short to ground at the EXT XMIT KEY input.

3.6.2.3 Audio (AUDIO).

3.6.2.3.1 Signal frequency (AUDIO). The AUDIO output signal frequency shall be in accordance with the response curves shown on Figure 2 when terminated in a 600 ohm resistive load.

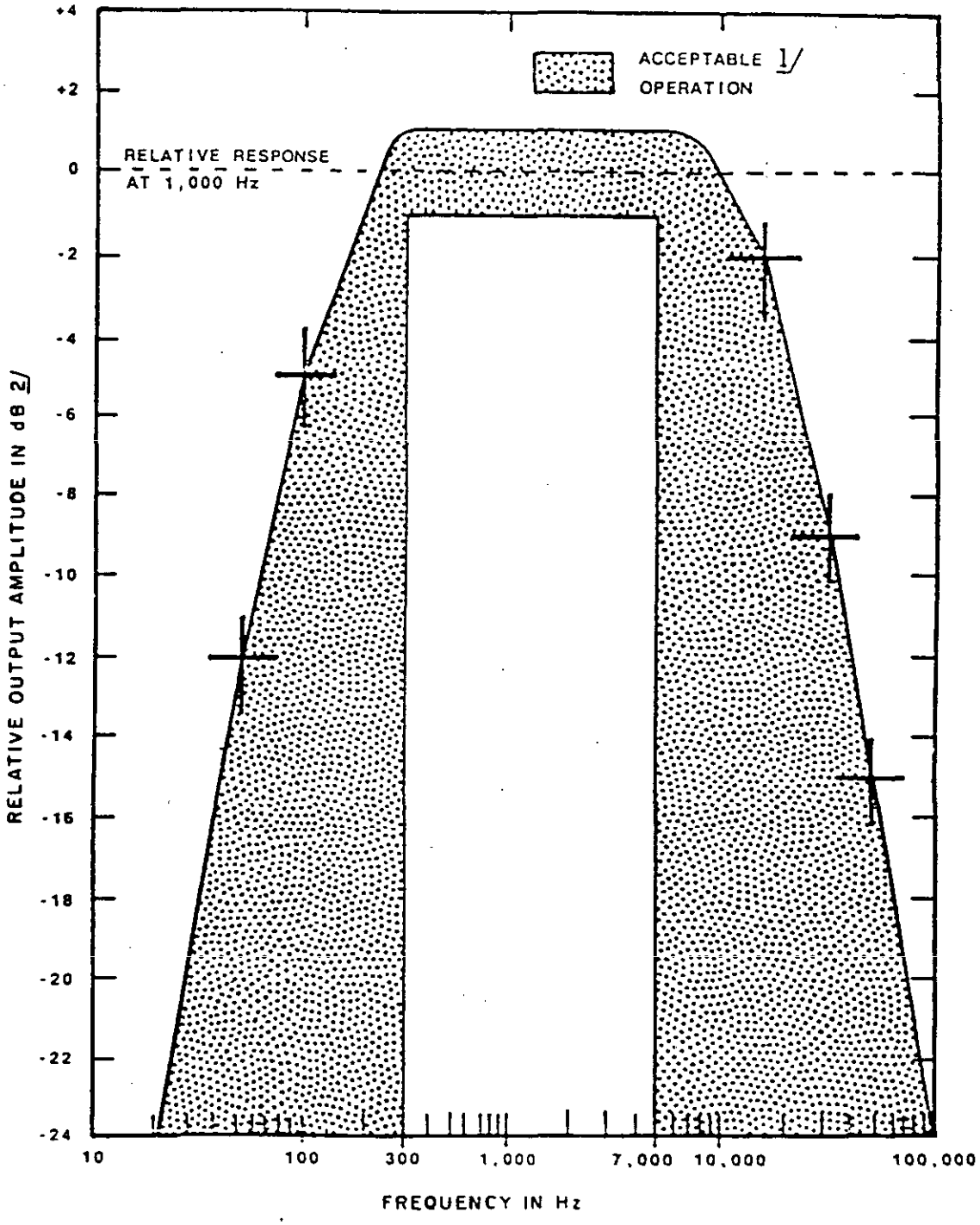
3.6.2.3.2 Signal amplitude/gain (AUDIO). The EMER RCVR NO. 1 and EMER RCVR NO. 2 signals shall be routed to the AUDIO output without attenuation when EMERG is selected on the AMPL SEL control. The EMER RCV NO. 3 signal shall be routed to the AUDIO output in accordance with the requirement of 3.6.1.9. For all other AMPL SEL switch selections, the AUDIO output shall be greater than nominal input (.06 volt at 1000 Hz applied to the SEL RCVR input) by 44 + 1 dB (8.48 - 10.67 V). The input/output characteristic (gain) of the amplifier shall be flat within +1 dB when the input level is varied between 0.01 and 0.10 volts rms. When the AMPL SEL switch is set to the NORM position, the AUDIO output shall also be an output when a signal is applied to the INTPH, AUX INTPH or MIC AUDIO inputs. The INTPH and AUX INTPH inputs shall be applied to a resistive divider network that includes the INTPH VOL control. The resultant AUDIO output shall be subject to amplitude variations in the range of 35 + 3 dB according to adjustment level. With the INTPH VOL control set fully clockwise, the AUDIO output shall be 44 + 1 dB (8.48 - 10.67 V) above an input of 0.06 Vac at 1000 Hz at either the INTPH or AUX INTPH input. When the INTPH VOL control is set fully clockwise, the interphone relay energized, and the microphone audio input is sufficient in amplitude to cause the INTPH output to be 7.0 VAC, the AUDIO output (sidetone) level shall be 0 $\pm$ 2 dB with respect to the 7.0 VAC INTPH output.

3.6.2.3.3 Distortion (AUDIO). The total harmonic distortion (with the AMPL SEL switch set to NORM, ALT 1 or ALT 2) over frequency range of 300 to 7000 Hz and input amplitude range of 0.01 to 0.10 V shall be not greater than 5 percent under standard conditions. The total harmonic distortion over the frequency range of 300 to 7000 Hz shall be not greater than 8 percent throughout the environmental envelope of MIL-E-5400 for Class 2 equipment when operated at rated power.

3.6.2.3.4 Output power (AUDIO). The output of the C-2645D shall be capable of delivering 1 watt into a 600 ohm resistive load.

3.6.2.3.5 Output stability. The output shall not show evidence of regeneration or oscillation throughout the environmental envelope of MIL-E-5400 for Class 2 equipment under any combination of input-output open, short, or complex impedance terminations. The C-2645D shall be capable of

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1/ Termination into 600 ohms or 7.5 ohms.
2/ Output amplitude is measured with respect to the amplitude at 1000 Hz.

FIGURE 2. Audio output signal response curves.

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withstanding any combination of open/short circuit terminations without damage and resume operation within specified requirements when properly terminated.

3.6.2.3.6 Internal noise. The C-2645D shall have an output of less than 5 millivolts when operated with the input shorted to signal ground throughout the environmental of MIL-E-5400 for Class 2 equipment.

3.6.2.4 Split (SPLIT). Split headset operation is not available during emergency operation. The SPLIT output signal shall be an open circuit whenever EMERG is selected on the AMPL SEL switch or conditions requiring interphone transmission exist resulting in actuation of the interphone relay. The input will consist of the input combinations specified in Table V. The output characteristics for the SPLIT operation are the same as those specified for the AUDIO output of 3.6.2.3 *except no volume control is provided.*

TABLE V. Input combinations.

Position	Input
EMERG	None
NORM	SPLIT
ALT 1	SEL RCVR, INTPH and AUX INTPH
ALT 2	SPLIT, INTPH, AUX INTPH and SEL RCVR

3.6.2.5 Transmit audio (XMIT AUDIO). *The XMIT AUDIO output shall be* obtained from the MIC AUDIO input after being routed through relay switching circuitry such that the output signal is identical to the input signal whenever a transmitter key command (ground) is received on the EXT XMIT KEY input.

3.6.2.6 Interphone (INTPH).

3.6.2.6.1 Signal frequency (INTPH). *The INTPH output signal amplitude* shall be in accordance with the response curves shown on Figure 2 when terminated in a 600 ohm non-inductive load except for the response from 300 to 7000 Hz. The response within this bandwidth shall be ±2 dB in lieu of ±1 dB.

3.6.2.6.2 Signal amplitude/gain (INTPH). The output shall be greater than a nominal input of 0.06 V at 1000 Hz applied to MIC AUDIO input by 39 ± 2 dB (4.24 - 6.73 V) *when terminated in a 600 ohm load. The input/output* characteristic (gain) of the amplifier shall be flat within ±1 dB when the input level is varied between 0.01 and 0.10 volts rms.

3.6.2.6.3 Output power (INTPH). The output of the amplifier shall be capable of delivering 1 watt into a 600 ohm resistive load.

3.6.2.6.4 Distortion (INTPH). The total harmonic distortion (with the AMPL SEL switch set to NORM, ALT 1 or ALT 2), over the frequency range of 300 to 7000 Hz and input amplitude range of 0.01 to 0.10 V, shall be not greater than 5 percent under standard conditions. The total harmonic distortion over

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the frequency range of 300 to 7000 Hz shall be not greater than 8 percent throughout the environmental envelope of MIL-E-5400 for Class 2 equipment when operated at rated power.

3.6.2.6.5 Output stability (INTPH). The output shall not show evidence of regeneration or oscillation throughout the environmental envelope specified herein under any combination of input-output open, short or complex impedance terminations. The C-2645D shall be capable of withstanding any combination of open/short circuit terminations without damage and resume operation within specified requirements when properly terminated.

3.6.2.6.6 Isolation (INTPH). When the microphone input circuit is properly terminated but no microphone signals are supplied, any signal appearing on the properly terminated outgoing interphone line shall be considered an undesired signal. The undesired signal level on the outgoing interphone line shall be at least 60 dB below rated power output. The undesired signal level on the outgoing interphone line shall be measured with the isolation amplifier properly terminated and driven to rated power output by either an incoming interphone or radio signal. The INTPH output, with the VOX actuated shall be suppressed a minimum of 60 dB.

3.6.2.7 Low impedance headset (HDST LOW Z).

3.6.2.7.1 Signal frequency (HDST LOW Z). The HDST LOW Z output shall be in accordance with the response curves shown on Figure 2 when terminated in a 7.5 ohm load.

3.6.2.7.2 Signal amplitude/gain (HDST LOW Z). The output shall be greater than a nominal input at SEL RCVR input of 0.06 V at 1000 Hz by 24.5 ± 1 dB (.90 - 1.13 V) when terminated in a 7.5 ohm resistive load (with the AMPL SEL switch in the NORM, ALT 1 or ALT 2 position). The input/output characteristic (gain) of the amplifier shall be flat within ±1 dB when the input level is varied between 0.01 and 0.10 volts rms.

3.6.2.7.3 Output power (HDST LOW Z). The output of the amplifier shall be capable of delivering 1 watt into a 7.5 ohm resistive load.

3.6.2.7.4 Distortion (HDST LOW Z). The total harmonic distortion over the frequency range of 300 to 7000 Hz shall be not greater than 5 percent under standard conditions. The total harmonic distortion over the frequency range of 300 to 7000 Hz shall be not greater than 10 percent throughout the environmental envelope of MIL-E-5400 for Class 2 equipment.

3.6.2.7.5 Output stability (HDST LOW Z). The output shall not show evidence of regeneration or oscillation throughout the environmental envelope specified herein under any combination of input-output open, short or complex impedance terminations. The C-2645D shall be capable of withstanding any combination of open/short circuit terminations without damage and resume operation within specified requirements when properly terminated.

3.6.2.7.6 Auxiliary microphone current (AUX MIC CUR). The AUX MIC CUR output, into a 100 ohm load connected to system ground, shall be 60 ± 10 milliamperes.

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3.6.2.7.7 +28 Vdc filtered power (FILT PWR). The FILT PWR output shall provide an output of 25.5 to 28.0 Vdc. The ripple present on the FILTERED POWER output shall be at least 35 dB below that present on the +27.5 Vdc input. The FILT PWR output shall exhibit the above characteristics under output current conditions from 0 to 250 milliamperes. The C-2645D shall be capable of sustaining a 250 milliampere FILT PWR output indefinitely.

3.6.3 Lighting.

3.6.3.1 Plastic lighting plates. Plastic lighting plate configurations shall meet the requirements of MIL-P-7788. The plastic lighting plate configuration shall be as specified in the contract or purchase order (see 6.2.1).

3.6.3.2 Front panel lighting. The front panel lamps and filters shall be as specified in the contract or purchase order (see 6.2.1).

3.7 SRAs. The C-2645D shall contain the SRAs listed in Table VI. Refer to detail specifications referenced in the table for the particulars of individual SRAs.

TABLE VI. SRA list.

Title	Application	Detail Specification
Amplifier	A 4501, A 4502	MIL-I-85845/1
VOX Module	A 4503	MIL-I-85845/2
Filter Module	FL 4501	MIL-I-85845/3

3.7.1 Amplifier. An amplifier module which provides 44 ± 1 dB of voltage gain shall be used in two places:

a. Interphone amplifier.

b. Isolation amplifier.

Performance shall be in accordance with MIL-I-85845/1.

3.7.2 VOX module. The VOX module shall provide a keyswitch function connecting the microphone output to the interphone line. Performance shall be in accordance with MIL-I-85845/2.

3.7.3 Filter module. This module shall provide filtering of the +28 Vdc supply used to power the VOX module, supply microphone and auxiliary microphone current. Performance shall be in accordance with MIL-I-85845/3.

3.8 Workmanship. Workmanship shall conform to MIL-STD-454, Requirement 9, and the following:

a. Printed wiring board coatings shall be free of dirt and metal particles.

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b. Chemical coated and painted surfaces shall have uniformly coated surfaces.

c. Metal parts shall be free of burrs or sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment and inspection facility. The contractor shall ensure that inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of the required inspections. Inspection equipment used in the inspections specified herein shall be maintained in calibration by a facility acceptable to the procuring activity which meets the requirements of MIL-STD-45662.

4.1.3 Quality assurance terms. Quality assurance terms used in this specification shall be as specified in MIL-STD-109.

4.1.4 Classification of inspections. The inspection requirements specified herein are classified as:

a. First article inspections (see 4.2).

b. Quality conformance inspections (see 4.3).

c. Special inspections (see 4.4).

4.1.5 Inspection conditions. Unless otherwise specified in the contract, all inspections shall be performed in accordance with the test conditions specified in 3.5.10.

4.2 First article inspection. First article inspections specified in 3.3 shall be conducted by the contractor on not less than one C-2645D representative of the processes, methods and techniques used on the C-2645Ds under

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contract. The first article inspection shall include all inspections required to verify compliance with all electrical and environmental requirements of this specification. The Government quality assurance representative and the procuring activity shall be advised as to when inspections are to be conducted so as to allow monitoring if desired. Contractors not having adequate facilities to conduct all the required inspections may obtain the services of a commercial inspection laboratory only when said laboratory is acceptable to the procuring activity. The procuring activity reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that the C-2645Ds conform to specified requirements. First article inspection shall include the following tests:

a. Altitude (see 4.2.1.1).

b. High temperature (see 4.2.1.2).

c. Low temperature (see 4.2.1.3).

d. Humidity (see 4.2.1.4).

e. Salt fog (see 4.2.1.5).

f. Vibration (see 4.2.1.6).

g. Mechanical shock (see 4.2.1.7).

h. Electromagnetic interference (see 4.2.2).

i. Electrical requirements inspection (see 4.2.3).

4.2.1 Environmental tests. Environmental tests shall be in accordance with MIL-STD-810, except as modified herein. The C-2645D shall operate as required herein during (unless as otherwise indicated) and after each environmental test. The C-2645D shall not sustain either physical damage or performance degradation during or as a result of exposure to the tests. The C-2645D shall be allowed a ±3 dB gain (see 3.6.2.3.2 and 3.6.2.6.2) when subjected to hot/cold temperature cycling. When the unit is returned to standard conditions, the gain shall remain ±1 dB for AUDIO and ±2 dB for INTPH.

4.2.1.1 Altitude. The C-2645D shall be subjected to an altitude test in accordance with MIL-STD-810, Method 500.2, Procedure II. The test altitude shall be limited to 50,000 feet and the rate of change shall be not greater than 2000 feet per minute.

4.2.1.2 High temperature. The C-2645D shall be subjected to high temperature operating test in accordance with MIL-STD-810, Method 501.2, Procedure II, except steps 3 through 6 of the procedure shall be deleted. The chamber temperature for step 7 shall be +71°C.

4.2.1.3 Low temperature. The C-2645D shall be subjected to low temperature operating tests in accordance with MIL-STD-810, Method 502.2, Procedure II. The chamber temperature shall be -54°C.

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4.2.1.4 Humidity. The C-2645D shall be subjected to humidity test in accordance with MIL-STD-810, Method 507.2, Procedure III, Figure 507.2-3. The test duration shall be 10 cycles.

4.2.1.5 Salt fog. The C-2645D shall be subjected to salt fog test in accordance with MIL-STD-810, Method 509.2, Procedure I. The test duration shall be 48 hours. Following the 48-hour drying period, the C-2645D shall meet performance requirements specified herein.

4.2.1.6 Vibration. The C-2645D shall be subjected to random vibration exposure in accordance with MIL-STD-810, Method 514.3, Procedure I for Category 5 equipment. Both functional and endurance tests shall be conducted using the vibration spectra shown on Figure 3. Vibration shall be applied sequentially along each of three mutually perpendicular axes for one-half hour at the functional level, and one hour at the endurance level followed by one-half hour at the functional level. The C-2645D shall operate as required during exposure to the functional level and immediately before and after exposure to the endurance level.

4.2.1.7 Shock. The C-2645D shall be subjected to shock tests in accordance with MIL-STD-810.

4.2.1.7.1 Functional. The C-2645D shall withstand three shocks in each direction along three mutually perpendicular axes in accordance with MIL-STD-810, Method 516.3, Procedure I and Figure 516.3-4 for operational test.

4.2.1.7.2 Crash hazard. The C-2645D shall be subjected to two shocks in each direction along three mutually perpendicular axes in accordance with MIL-STD-810, Method 516.3, Procedure V and Figure 516.3-4 for crash hazard test. There shall be no failure of the mounting attachment and the C-2645D shall remain in place and not create a hazard. Bending and distortion shall be permitted.

4.2.1.7.3 Bench handling. The C-2645D shall be tested in accordance with MIL-STD-810, Method 516.3, Procedure VI.

4.2.2 Electromagnetic interference (EMI). The C-2645D shall be subjected to the EMI tests specified in 4.2.2.1 through 4.2.2.2.6 in accordance with the test methods of MIL-STD-462 to verify that C-2645D meets the requirements of MIL-STD-461, Class A1, Category A1b, for the EMI tests herein.

4.2.2.1 Conducted and radiated emission.

4.2.2.1.1 Conducted emissions, power and interconnecting leads, 0.15 to 50 MHz. Conducted emissions measurements shall be made over the frequency range of 0.15 to 50 MHz on power leads in accordance with MIL-STD-462, Test Method CE03, and interconnecting leads in accordance with MIL-STD-462, Test Method CE04.

4.2.2.1.2 Conducted emissions power leads, spikes, time domain. The contractor shall devise a test to demonstrate compliance with the requirements of 3.5.14.1.2.

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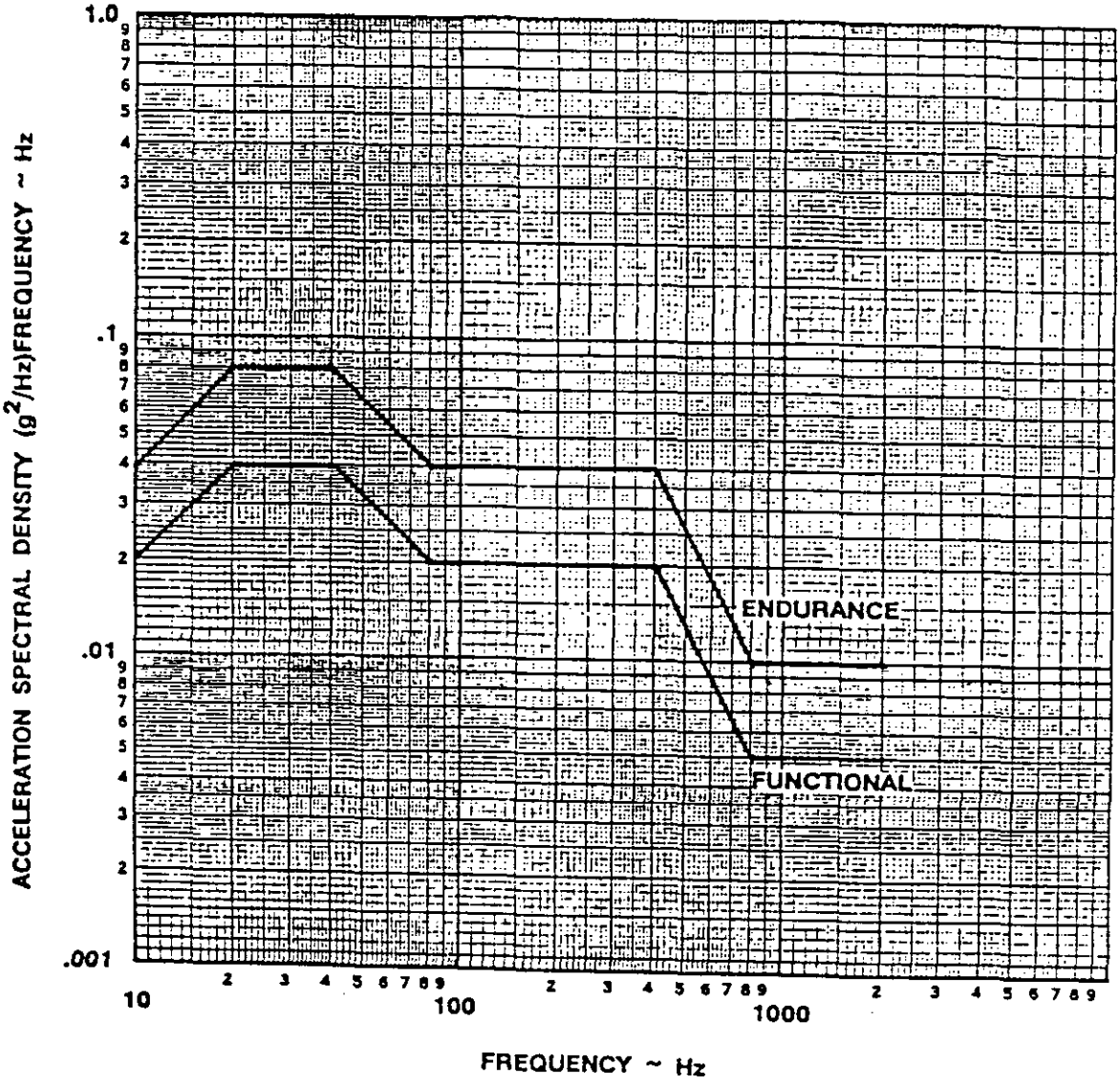


FIGURE 3. Vibration test spectrum.

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4.2.2.1.3 Radiated emissions, magnetic field, 0.03 to 50 kHz. Radiated emissions emanating from the C-2645D shall be measured over frequency range of 0.03 to 50 kHz (magnetic field) in accordance with MIL-STD-462, Test Method RE02.

4.2.2.2 Conducted and radiated susceptibility.

4.2.2.2.1 Conducted susceptibility, power leads, 30 Hz to 50 kHz. The C-2645D shall be subjected to conducted susceptibility tests by having signals over the range of 30 Hz to 50 kHz injected on the power leads in accordance with MIL-STD-462, Test Method CS01. Susceptibility shall be determined in accordance with criteria of 3.5.14.3.2.

4.2.2.2.2 Conducted susceptibility, power leads, 50 kHz to 400 MHz. The C-2645D shall be subjected to conducted susceptibility tests by having signals over the range of 50 kHz to 400 MHz injected on the power leads in accordance with MIL-STD-462, Test Method CS02. Susceptibility shall be determined in accordance with criteria of 3.5.14.3.2.

4.2.2.2.3 Conducted susceptibility, spikes, power leads. The C-2645D shall be subjected to conducted susceptibility tests consisting of superimposing a 32 volt maximum spike on the power line voltage waveform in accordance with MIL-STD-462, Test Method CS06. Susceptibility shall be determined in accordance with criteria of 3.5.14.3.2.

4.2.2.2.4 Radiated susceptibility magnetic induction field spikes and power frequencies. The C-2645D shall be subjected to radiated susceptibility tests consisting of spikes and power frequencies in accordance with MIL-STD-462, Test Method RS02. Susceptibility shall be determined in accordance with criteria of 3.5.14.3.2.

4.2.2.2.5 Radiated susceptibility, electric field, 14 kHz to 40 GHz. The C-2645D shall be subjected to radiated susceptibility tests over the frequency range specified in Table I, electric field, in accordance with MIL-STD-462, Test Method RS03.

4.2.2.2.6 Ground plane interference. The contractor shall devise a test to demonstrate compliance with the requirements of 3.5.14.4. This test shall be included as a part of the EMI test plan.

4.2.3 Electrical requirements inspection. The contractor shall conduct inspections necessary to demonstrate compliance to the electrical requirements of Section 3 and DL 1794AS7000. Demonstrations shall include but not be limited to the tests herein.

4.2.3.1 Warm-up time. Time required for the C-2645D to stabilize after being energized at standard and environmental conditions shall be monitored to assure that the C-2645D meets the requirements of 3.5.12.

4.2.3.2 Input power.

4.2.3.2.1 Steady state power.

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4.2.3.2.1.1 Normal steady state. The C-2645D shall be checked operationally with input voltage set at 22 volts dc and 29 volts dc in accordance with MIL-STD-704 for normal steady state test. The C-2645D shall be operated long enough to assure proper operation with each input voltage.

4.2.3.2.1.2 Emergency steady state. The C-2645D shall be checked operationally with input voltage set at 16 volts dc and 29 volts dc in accordance with MIL-STD-704 for emergency steady state test. The C-2645D shall be operated long enough to assure operation with each input voltage.

4.2.3.2.2 Transient testing.

4.2.3.2.2.1 Voltage surge. The C-2645D shall be subjected to input voltage surges of 18 to 50 volts dc for 0.013 to 0.042 second (transient duration) in accordance with MIL-STD-704 for transient test. The C-2645D operation shall be observed for permanent degradation.

4.2.3.2.3 Overvoltage, undervoltage and power interruption.

4.2.3.2.3.1 Overvoltage. The C-2645D shall be subjected to input voltage of 50 volts dc for 0.045 second in accordance with MIL-STD-704 for overvoltage test. The C-2645D operation shall be observed for permanent degradation.

4.2.3.2.3.2 Undervoltage. The C-2645D shall be subjected to a change of input voltage from normal to 0 volt dc for 5 seconds in accordance with MIL-STD-704 for undervoltage test. The C-2645D operation shall be observed for permanent degradation.

4.2.3.2.3.3 Power interruption. The C-2645D shall be subjected to input power interruption of 50 ms in accordance with MIL-STD-704 for power interruption test. The C-2645D operation shall be observed for permanent degradation.

4.3 *Quality conformance inspection.*

4.3.1 Individual inspections. Each production C-2645D submitted for acceptance shall be subjected to the required inspections specified herein. These inspections shall be adequate to determine compliance with the requirements for material, workmanship, operational performance and environmental stress screening (ESS).

a. Examination of product (see 4.3.1.1).

b. Operational inspection (see 4.3.1.2).

c. Environmental stress screening (ESS) (see 4.3.1.3).

d. Production reliability acceptance test (PRAT) (see 4.3.1.4).

4.3.1.1 Examination of product. Each production C-2645D shall be examined carefully to ensure compliance with requirements of workmanship, marking, interchangeability, design and construction.

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4.3.1.2 Operational inspections. Each production C-2645D shall be subjected to an operational inspection. The inspection shall demonstrate and test all modes of operation; all input/output signals shall be tested and the results recorded. During the inspection sufficient characteristics shall be measured and recorded to assure that the C-2645Ds submitted for acceptance meet all the requirements of this specification.

4.3.1.2.1 Input impedance. Verify that the input impedances are in accordance with 3.6.1.1.3, 3.6.1.2.3 and 3.6.1.3.3 herein. Impedances shall be measured at 1000 Hz. The input impedance shall be measured using the resistance substitution method.

4.3.1.2.2 Gain. Verify that the amplifier gain at 1000 Hz is in accordance with 3.6.2.3.2, 3.6.2.6.2 and 3.6.2.7.2.

4.3.1.2.3 Frequency response. Holding the input signal amplitude constant, vary the frequency between 300 - 7000 Hz. Verify that the amplifier gain meets the requirements of 3.6.2.3.1, 3.6.2.6.1 and 3.6.2.7.1.

4.3.1.2.4 Linearity. Using a 1000 Hz input signal which is varied between 0.01 Vac and 0.1 Vac, verify that the amplifier meets the requirements of 3.6.2.3.2, 3.6.2.6.2 and 3.6.2.7.2.

4.3.1.2.5 Output power. With the line out properly terminated, apply a 1000 Hz signal of appropriate amplitude to verify the requirements of 3.6.2.3.4, 3.6.2.6.3 and 3.6.2.7.3.

4.3.1.2.6 Distortion. Verify the requirements of 3.6.2.3.3, 3.6.2.6.4 and 3.6.2.7.4.

4.3.1.2.7 Stability. Verify the requirements of 3.6.2.3.5, 3.6.2.6.6 and 3.6.2.7.6.

4.3.1.2.8 Internal noise. Verify the requirements of 3.6.2.3.6.

4.3.1.2.9 Susceptibility to stray magnetic fields. Verify the performance of 3.5.14.3.2.

4.3.1.2.10 Controls. Verify control operation of 3.5.15.2.

4.3.1.2.11 Audio susceptibility. Verify the resistance to audio interference of 3.5.14.3.2 by conducting tests in accordance with MIL-STD-461.

4.3.1.3 Environmental stress screening (burn-in). An environmental stress screening inspection shall be required on all production C-2645Ds. Prior to and after each submission to the screening inspection a full electrical performance inspection under standard conditions shall be performed on each production C-2645D. Each production C-2645D shall be subjected to a workmanship random vibration of 6.3 g's rms with a frequency of 20 to 2000 Hz, as measured on the vibration fixture at the mounting points (see Figure 4). The vibration duration shall be not less than 10 minutes (failure free) in the axis that is perpendicular to the majority of the printed wiring boards. The production C-2645D shall be mounted to the vibration fixture by its normal

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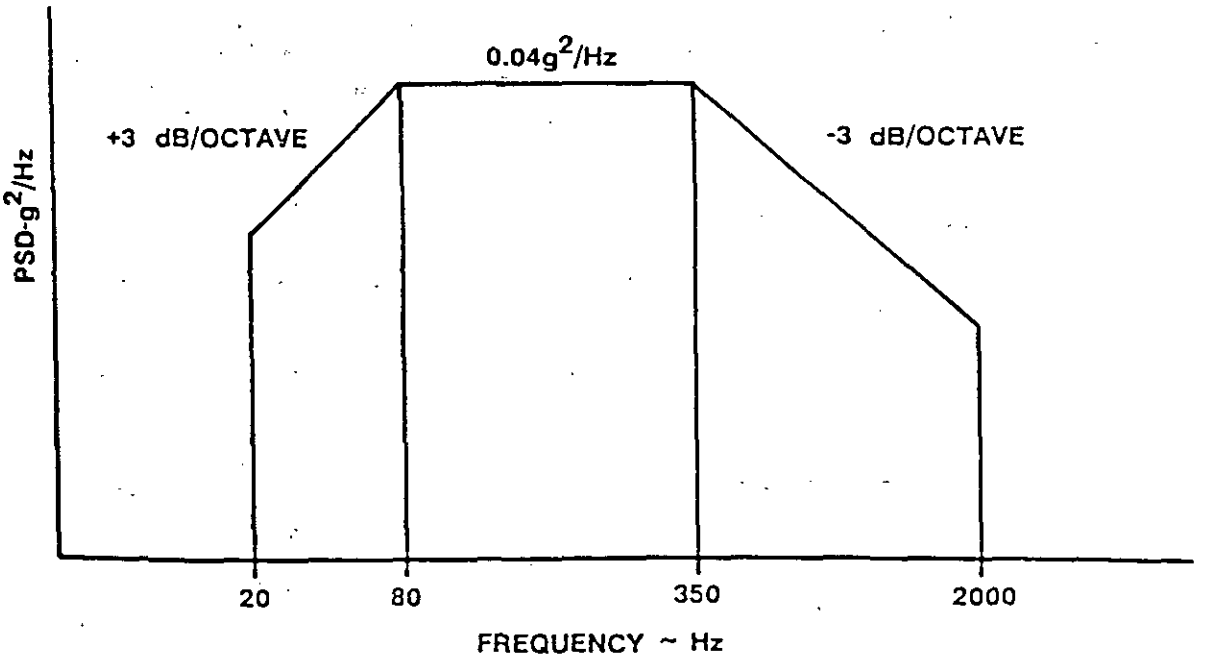


FIGURE 4. Workmanship random vibration.

mounting means so as to meet the vibration requirements specified herein. The production C-2645D shall be continuously energized and electrically monitored for the duration of the inspection. After the vibration inspection each production C-2645D shall be subjected to not less than 72 hours of thermal stress screening of which, not less than 45 hours shall be with the C-2645D energized. The thermal stress screening inspection shall consist of nine 8-hour temperature cycles of which each cycle contains a 3-hour cold period at -54°C (C-2645D deenergized) and a 5-hour hot period at +71°C (C-2645D energized). Each production C-2645D shall be deenergized for the duration of the cold period except for the time needed to perform the electrical performance inspection specified herein (see Figure 5).

Each production C-2645D shall be energized for the full duration of the hot period. An electrical performance inspection shall be performed once daily during both hot and cold periods, after the production C-2645Ds have stabilized at inspection temperatures. The last two temperature cycles shall be failure free. The thermal stress screening inspection chamber's temperature change rate shall be greater than 10°C per minute. The contractor shall be afforded the option of performing the vibration inspection concurrently with the temperature cycling portion of the screening inspection or as a stand-alone inspection.

4.3.1.4 Production reliability acceptance test (PRAT). Following the environmental stress screening inspection of 4.3.1.3 each C-2645D shall be subjected to a production reliability acceptance test as a means for verifying the MTBF requirement of 3.5.1.2 (1000 hours). Only those C-2645Ds that have

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PRODUCTION RELIABILITY ACCEPTANCE TEST
ENVIRONMENTAL TEMPERATURE CYCLE
C-2645D/AIC-14A

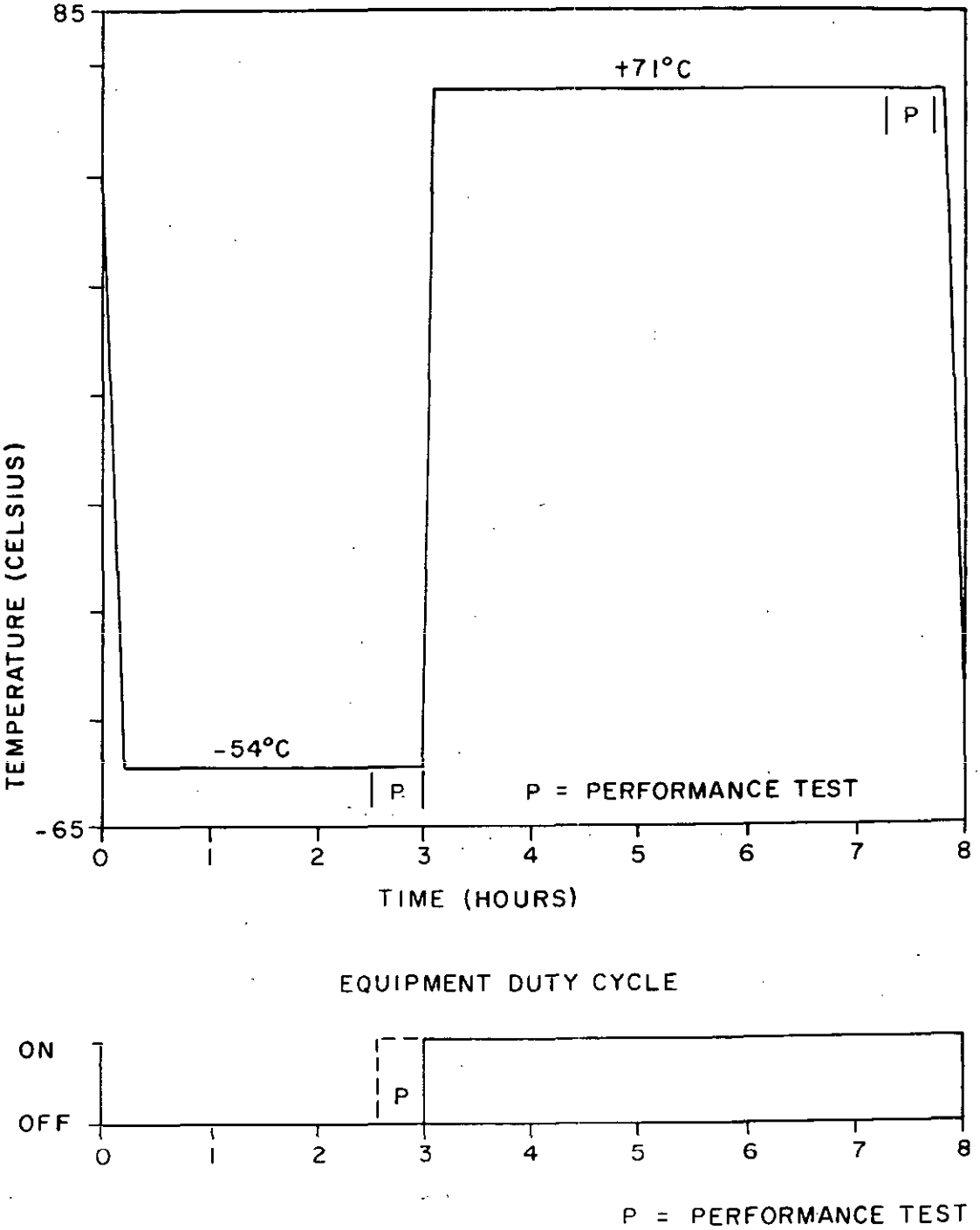


FIGURE 5. Environmental cycling conditions.

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successfully completed both the environmental stress screening inspection and the production reliability acceptance test shall be delivered to the procuring activity. The production reliability acceptance test shall be in accordance with MIL-STD-781, Test Plan XVIIIC. Each C-2645D shall accumulate a minimum of 45 operating hours (last three cycles failure free) under environmental cycling conditions as shown on Figure 5. No particular C-2645D shall be allowed to accumulate more than 90 hours of production reliability acceptance test time.

4.3.1.5 Determination of compliance. The total production reliability acceptance test time and chargeable failures will be plotted on the Continue Test/Reject Criteria Graph of MIL-STD-781, Test Plan XVIIIC, using a lower test MTBF O_1 of 1000 hours. The upper test MTBF O_0 shall be 2000 hours (see Figure 6). If the plot of test failures versus total test time enters the reject region, testing shall be suspended and the procuring activity along with the quality assurance representative shall be notified.

4.4 Special inspections. The procuring activity reserves the right to perform any one or group of inspections set forth in this specification when such inspections are deemed necessary by the procuring activity to assure production item conformance to the specified requirements. Conditions that warrant special inspections shall include:

a. First production C-2645D incorporating a design or material change.

b. Whenever production is interrupted for more than 1 year.

c. Whenever failure reports or other information indicate that additional inspections are required.

d. Whenever it is deemed necessary to assure production C-2645Ds are equal to the previously approved first article inspection C-2645D in design, construction, workmanship, performance and quality.

e. As specified in the contract or purchase order (see 6.2.1).

4.5 Classification of failures. Should a failure occur during any inspection, the following action shall be taken:

a. Determine the cause of failure.

b. Determine whether the failure is an isolated case or design or production process defect.

c. Implement corrective action intended to reduce the possibility of the same failure(s) occurring in future inspections.

d. Where practical, include additional sequences during the quality conformance acceptance inspections until assurance is obtained that the defect has been corrected.

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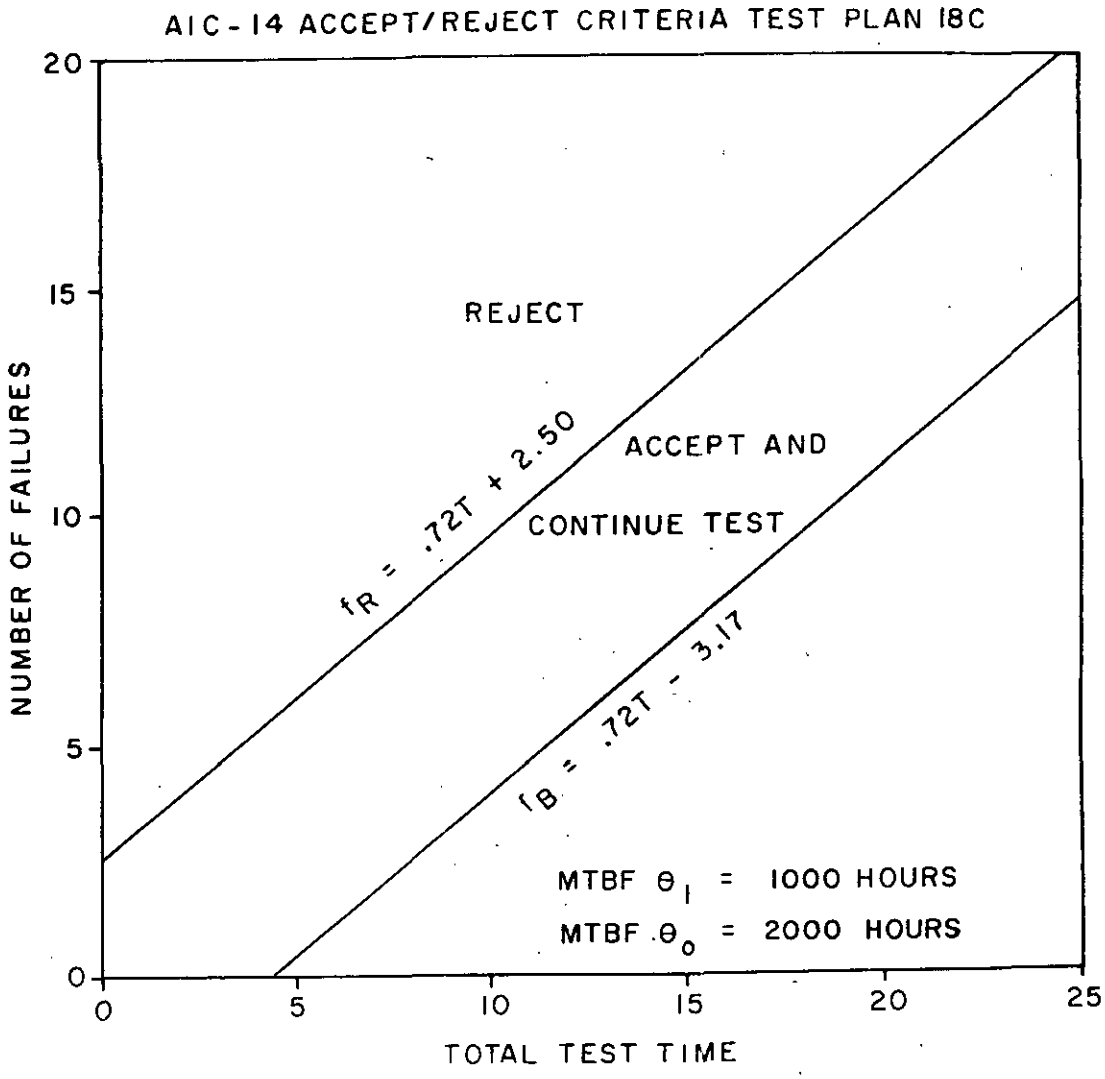


FIGURE 6. Continue test/reject criteria graph.

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4.6 Methods of examination and inspection. All inspections shall be conducted in accordance with Government approved procedures. When approved inspection procedures are available from previous contracts, they may be provided and used upon authorization of the procuring activity. The Government reserves the right to modify procedures or add inspections when it is deemed necessary to assure C-2645Ds are in compliance with this specification and the contract.

5. PACKAGING

5.1 General. Packaging shall be in accordance with the general requirements of MIL-E-17555, supplemented as follows.

5.2 Preservation-packaging. Unless otherwise specified in the contract or purchase order (see 6.2.1), preservation-packaging shall be in accordance with MIL-P-116, Method III.

5.3 Packing. Unless otherwise specified in the contract or purchase order (see 6.2.1) packing shall be level A. The packaged equipments shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, normal requirements. Sealing (taping) and reinforcing (banding) shall be in accordance with the appendix to PPP-B-636.

5.3.1 Unitized loads. Unitized loads, commensurate with the level of packing specified herein or in the contract (see 6.2.1) shall be used whenever the total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent possible.

5.4 Marking. In addition to any special marking required by the contract (see 6.2.1) each unit package, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The C-2645D Master Interphone Control Panel is intended to be a major component of the AN/AIC-14A Intercommunications Set.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Selection of plastic lighting plates (see 3.6.3.1).

c. Selection of panel lamps and bulbs (see 3.6.3.2).

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d. Selection of applicable levels of packaging and packing (see Section 5).

e. When a first article is required (see 4.2).

f. When special tests are required (see 4.4).

6.3 First article. When a first article inspection is required, the C-2645D will be tested and should be a first article sample. The first article should consist of two C-2645Ds. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first article samples. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (keyword) listing.

AN/AIC-14A
Intercommunications
Master Interphone C-2645D Set
Panel, control

Preparing Activity:
Navy - AS
(Project No. 5831-N079)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I-85845A (AS)	14 NOVEMBER 1990

3. DOCUMENT TITLE INTERCOMMUNICATIONS SETS; AN/AIC-14A MASTER INTERPHONE CONTROL PANEL C-2645D/AIC-14 GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDING OFFICER NAVAL AIR ENGINEERING CENTER SYSTEMS ENGINEERING & STANDARDIZATION DEPT	b. TELEPHONE (Include Area Code) (1) Commercial: (908)323-2326 (2) AUTOVON: 624-2326
c. ADDRESS (Include Zip Code) CODE 53 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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